

Literature Review on Sensor Technologies for Wastewater Collection Networks: Operating Principles, Field Failure Modes, and Deployment Trade-Offs

August 21, 2026

Abstract

This review classifies wastewater collection-network monitoring by the question each technology answers: how full the sewer is, how much is flowing, what is in the wastewater, whether the asset is damaged or blocked, and how measurements are transmitted. The review was rewritten using an evidence-bounded audit of eleven full-text research papers and selected publication records. The evidence supports non-contact radar as a promising level-monitoring option in humid sewer environments, but it does not support reporting a generic 77–81 GHz field deployment, a fifteen-manhole twelve-month trial, or a universal radar accuracy or battery-life value from Drenoyanis et al. [6]. The frequently quoted 72.14% flow error in Alshami et al. [1] is retained only with its correct modality and context: contactless ultrasonic sensors at twelve Hong Kong locations compared with Manning-equation estimates. Optical and water-quality measurements are treated as calibrated proxies or event indicators, while gas-phase and liquid-phase H₂S monitoring are kept as distinct categories. Procurement costs, lifetimes, maintenance intervals, and deployment percentages are not generalised unless the cited source actually reports them.

Keywords: wastewater collection; sewer monitoring; level sensing; flow sensing; water quality; optical spectroscopy; hydrogen sulfide; sensor placement.

1 Scope, Method, and Evidence Rule

This paper concerns fixed or remotely operated monitoring in wastewater collection networks. It is a literature review, not a product quotation, a substitute for SA Water asset records, or a validated deployment design. The classification used here is an organising framework for this project rather than a universal taxonomy.

The evidence audit used the full-text papers stored with the project, including studies of radar and Lo-RaWAN, low-power distance sensing, contactless flow sensing, Melbourne discharge detection, self-cleaning water-quality monitoring, fluorescence, UV/Vis calibration, and gas- and liquid-phase H₂S monitoring. Each claim was checked against the source’s measurement modality, test environment, sample size, result type, and stated limitation. A result from a prototype, a manufacturer’s specification, or a future-work proposal is not presented as a field-validated network result.

The following rule is applied throughout:

A numerical claim is included only when the source reports that value for the same device, environment, and purpose. Otherwise the claim is labelled as a proposal, a device-specific specification, an inference, or an unverified project input.

2 Monitoring Questions and Sensor Categories

The literature is clearer when grouped by the question being answered rather than by a single list of hardware names.

Table 1: Question-led classification of sewer monitoring technologies.

Question	Examples	Information obtained	Main limitation
How full is the sewer?	Radar, ultrasonic, short-range LiDAR/ToF, float or tilt switches	Continuous level trends or discrete surcharge events	Humidity, condensation, foam, obstructions, fouling, or loss of trend information
How much is flowing?	Contactless ultrasonic flow, level plus geometry, pump drawdown or hydraulic models	Flow or velocity estimates and hydraulic loading	Installation geometry, flow direction, backwater, and calibration assumptions
What is in the wastewater?	pH, conductivity, temperature, turbidity, UV/Vis, fluorescence, H ₂ S	Chemical changes, illicit-discharge indicators, organic-load proxies, odour or corrosion indicators	Fouling, matrix effects, local calibration, and phase-specific measurement
Is the asset blocked or damaged?	Acoustic screening, CCTV, computer vision, fibre-optic or other condition methods	Structural or blockage information	Usually an inspection or diagnostic workflow rather than a continuous level time series
How does the reading get back?	LoRaWAN, cellular IoT, gateways, data platforms, soft sensors	Remote transmission and estimation of unmeasured states	Coverage, power, data quality, model dependence, and site-specific validation

Telemetry and soft sensors are enabling layers, not additional physical water-quality or hydraulic sensor types. Likewise, acoustic and CCTV methods answer an asset-condition question and should not be presented as equivalent to a fixed level probe.

3 Level and Depth Sensing

Continuous level data provide a time series of depth from which a utility may examine normal patterns, surcharge, and rate of change. A binary float or tilt switch provides an event threshold but cannot produce the same trend information. The choice therefore depends on whether the monitoring purpose is broad event notification, hydraulic diagnosis, or both.

3.1 Non-contact Radar

Radar level sensors measure the time or frequency response of an electromagnetic signal reflected from the wastewater surface. In practical terms, the sensor converts the reflected signal into a distance and then into a water-level measurement. The measurement principle does not establish the frequency, accuracy, or lifetime of a particular sewer installation.

Drenoyanis et al. [6] selected non-contact radar for sewer level and pumping-station drawdown measurements. Their paper describes two radar platforms: a portable sewer-pumping-station drawdown device and a LoRaWAN-enabled prototype. A prototype end device was installed in Shoalhaven Water’s network, and a separate communications survey assessed LoRaWAN performance. The paper reports an urban transmission survey reaching up to 20 km and line-of-sight distances up to 35 km, but these are communications results rather than evidence of a fifteen-manhole, twelve-month radar trial.

The paper’s comparison with ultrasonic sensing is narrower and more defensible than the original draft’s wording. It reports that steam, mist, condensation, foam, and air-temperature changes can create problems for ultrasonic measurement, while radar is less affected by humidity and mist. It also states that incorrectly installed radar can produce false positives and that radar is more expensive than alternatives. Radar should therefore be described as a promising non-contact option, not as immune to all sewer interference or as a universal highest-resilience technology.

The paper describes an integrated 80 GHz short-range radar network and approximately five devices around each pumping station as *future work* [6, pp. 28–29]. It does not establish a generic 77–81 GHz field range, sub-millimetre field accuracy, five-year operation, or a validated overflow-prediction model. These values and outcomes must not be attributed to the field results.

3.2 Ultrasonic, LiDAR, and Time-of-Flight Sensors

An ultrasonic level sensor estimates distance from the round-trip travel time of an acoustic pulse. Sound speed varies with environmental conditions. Drenoyanis et al. [6, pp. 3–4] identify steam, mist, water condensation, foam, and temperature-dependent sound speed as practical sources of false or unstable readings. The source

does not establish the earlier draft’s 5–12% error range, a 0.15–0.5 m dead-band change, or a universal beam-reflection mechanism involving step irons and benching.

Utepov et al. [16] evaluated Sharp infrared, TF-Luna LiDAR, TOF400 laser, and JSN-SR04T ultrasonic sensors in a prototype arrangement intended to imitate a sewer well. The reported test results are device-specific and not a field validation across diverse sewer systems. The paper reports, among other results, JSN-SR04T performance of approximately ± 5 cm up to 2 m and TF-Luna performance no worse than approximately ± 8 cm up to 5 m. It also reports standard-deviation ranges for its distance tests and states that field trials in different sewer systems remained future work. The complete prototype material cost was about USD 170.07 excluding labour; this is not a generic ultrasonic-node capital cost.

Short-range optical distance sensing may therefore be considered as a candidate technology, but laboratory or imitative accuracy must not be reported as sewer field accuracy. Condensation protection, optical fouling, mounting geometry, and validation in the target asset remain necessary.

3.3 Binary Float and Tilt Sentinels

Float and tilt devices answer a simpler question: has wastewater crossed a set threshold? Their binary output can be useful for distributed event notification, but it cannot calculate a rate of rise or describe the level between the trigger and reset thresholds. The reviewed research set does not establish universal prices, five-to-ten-year battery life, grease failure rates, or maintenance intervals for float or tilt devices. Those values require product-specific datasheets, procurement evidence, and SA Water maintenance records.

4 Flow and Velocity Measurement

For a partially full sewer, discharge depends on both the wetted cross-sectional area and the mean water velocity. A level sensor alone does not automatically provide a reliable flow measurement when the flow is non-uniform or the geometry and boundary conditions are uncertain.

4.1 Contactless Ultrasonic Flow: The Alshami Result

Alshami et al. [1] studied twelve sewer locations in Hong Kong using contactless ultrasonic flow and level sensors. They reported a level–flow correlation coefficient of 0.36 and an average relative flow-rate error of 72.14% when compared with Manning-equation estimates. The value is important, but its context must remain attached: it is not a radar result, not a universal error for contactless flow sensing, and not proof that every Manning calculation fails.

The authors discuss several possible causes of error, including small pipe diameters, installation angle, inconsistent flow direction, water level, and multiple inlets or outlets. The study therefore supports site-specific installation and calibration. It does not establish the earlier draft’s generic “uncalibrated radar” wording, a universal surface-velocity conversion factor, or a universal requirement for one particular calibration procedure.

4.2 Level-derived Flow and Manning Models

Manning-type level-to-flow estimates can be useful under their assumptions, but those assumptions should be checked against the target sewer reach. In a blocked or backwatered reach, the water level may rise while the velocity or discharge does not rise proportionally. The practical conclusion is that level-to-flow conversion requires hydraulic context, geometry, calibration, and validation against observed flow where possible. The 72.14% Alshami result illustrates this risk in one network and instrument configuration.

Drenoyanis et al. [6] also used radar level measurements for pumping-station drawdown and indirect pump-flow estimation. That is a different application from contactless flow sensing in a gravity sewer and should not be merged with the Alshami modality.

5 Water-Quality and Chemical Monitoring

Hydraulic sensors indicate how full the sewer is and how it behaves. Water-quality sensors can indicate what has entered the system, whether the water has been diluted, and whether chemical conditions create an odour, corrosion, or treatment concern. Continuous water-quality monitoring is not required at every manhole; it is most defensible at targeted locations where a chemical signal answers a defined operational question.

5.1 Physicochemical Event Detection

Shi et al. [14] deployed 22 low-cost level, temperature, and electrical-conductivity sensor units across an urban Melbourne catchment for more than one year. The study detected dry-weather illicit discharges and transient events. It did not use pH in the reported sensor set, so it should not be cited as evidence for a pH/EC/temperature node. It also does not provide a general “high event detection rate” without a defined numerator, denominator, and treatment of missed or false events.

Li et al. [8] evaluated a sewage-quality monitoring system using temperature, conductivity, turbidity, and pH measurements. The system included wall-jet cleaning and self-diagnostic features. The paper reports field evaluation over a broader three-year programme and continuous raw-sewage operation for up to 24 months without maintenance or failure at a reported trial site. This supports the value of self-cleaning design in that system; it does not establish a universal monthly calibration interval.

5.2 UV/Vis and Fluorescence

Lepot et al. [7] review UV/Vis methods for estimating TSS and total or dissolved COD. A central limitation is that a new site or water matrix generally requires local calibration and verification. The method should therefore be described as an optical proxy, not as universally accurate reagent-free COD measurement.

Carstea et al. [3] report that fluorescence features including tryptophan-like Peak T and humic-like Peak C can correlate with BOD, COD, or TOC under appropriate conditions. The review also reports overlapping fluorophores, inner-filter effects, turbidity, fouling, and variation between sample matrices and methods. The defensible wording is “can correlate after validated, site-specific calibration”, not “direct” or “instant” BOD measurement.

Optical instruments therefore provide useful rapid indicators, but the relevant cleaning method, calibration schedule, and performance limits must be specified for the actual instrument and wastewater matrix. Wipers, air flushing, and other cleaning approaches should not be presented as universal requirements without a device-specific source.

Other technologies, including nutrient ion-selective electrodes, dissolved-oxygen probes, and microbial fuel-cell biosensors, are valid research categories. They are not assigned prices, lifetimes, response times, or sewer-network performance values here because those values were not part of the verified evidence set used for this audit.

6 Hydrogen Sulfide and Corrosion Monitoring

Hydrogen sulfide is relevant to sewer odour, worker safety, and biogenic corrosion. The sensor phase matters: gas-phase headspace measurements and liquid-phase dissolved-sulfide measurements are different technologies and cannot be combined into one generic sensor category.

6.1 Gas-Phase Sensors

Pacheco Fernández et al. [10] compared three commercial gas sensors at two sewer sites. Their device table reports different ranges and operating specifications: OdaLog and MyData devices are listed at 0–200 ppm, while SulfiLogger is listed at 0–1000 ppm. The same table reports device-specific humidity and battery specifications, including approximately eight months, three months at a stated transmission configuration, and two years for the compared devices. These are manufacturer or configuration-specific values, not universal sewer lifetimes.

The study found that readings were more reliable at low concentrations and that deviations increased with concentration. It discusses high-humidity effects and device-management procedures, but it does not demonstrate that all gas-sensor cells degrade after 12–18 months at 100% relative humidity. A gas-phase range such as 0–200 ppm must therefore name the device.

6.2 Liquid-Phase Sensors

Despot et al. [4] studied online liquid-phase H₂S sensors, including UV/Vis and electrochemical instruments. The study reports dependence on local calibration, disturbance from solids, and the need for regular inspection and maintenance. It discusses critical monitoring points such as the end of a pressure sewer and locations downstream of dosing. It does not support a gas-sensor replacement interval, a universal six-month field life, or a quantified 35% chemical-consumption reduction.

Pandey et al. [11] provide a broader review of sensor-based H₂S monitoring. The review supports treating H₂S as a distinct monitoring problem, but it should not be used to attach an unreported crown-loss rate or a product-specific maintenance interval to every sewer installation.

7 Pipe Condition and Inspection Methods

Acoustic screening, CCTV, computer vision, and fibre-optic methods answer a different question from fixed hydraulic or chemical probes: whether a pipe or appurtenance has a structural, blockage, or condition problem. The project bibliography contains reviews and studies of these methods [2,9,12,13,17]. In this evidence-bounded rewrite, no unverified range, blockage percentage, inspection interval, or accuracy value is assigned to them. Any such value should be copied from the exact primary source and test configuration before formal submission.

8 Fouling, Maintenance, Telemetry, and Power

The reviewed papers repeatedly show that fouling, solids, condensation, calibration, and access are operational constraints. Li et al. [8] demonstrate a particular wall-jet cleaning design, while Despot et al. [4] report regular inspection and maintenance for liquid-phase H₂S sensing. These findings support including maintenance and calibration in the design problem; they do not justify applying one interval to every sensor type.

Non-contact measurement avoids direct immersion, but it does not guarantee zero maintenance. Radar can still be affected by incorrect installation, beam geometry, or deposits on exposed surfaces. Contacting and submerged devices face additional fouling and access issues, but the magnitude and timing are device- and site-specific.

Drenoyanis et al. [6] provide an Australian LoRaWAN communications case study and a regional signal survey. The reported transmission distances do not prove reliable penetration through every cast-iron or concrete cover. A target deployment requires a site-specific radio survey, antenna arrangement, gateway assessment, and battery/telemetry duty-cycle test. Sikorski et al. [15] formulate network coverage and cost decisions that include sensor, battery, and sampling considerations, but the model does not establish a universal five-to-ten-year battery life or a radar-versus-float cost ranking.

Accordingly, this paper does not state generic values for RF attenuation, battery life, power draw, capital cost, annual maintenance, or total cost of ownership. Those values belong in a separate procurement and field-validation register.

9 Evidence-Bounded Comparative Synthesis

Table 2: What can and cannot be concluded from the audited evidence.

Modality	Primary output	Defensible advantage	Important limitation	Evidence anchor
Radar level	Continuous level; indirect pump-flow information in a defined application	Non-contact measurement and lower sensitivity than ultrasonic to humidity and mist in the Drenoyanis comparison	Setup, beam geometry, cost, and field validation remain important	Drenoyanis, pp. 1, 3–4, 18–19, 28–29
Ultrasonic level	Continuous distance or level	Practical low-power distance sensing and demonstrated use in a South Australian early-choke study	Humidity, condensation, foam, temperature, and installation conditions affect readings	Drenoyanis, pp. 3–4; Do et al. [5]
LiDAR/ToF prototype	Distance in a test arrangement	Candidate low-power distance modalities	Utepov’s arrangement was imitative; diverse sewer field validation was future work	Utepov, pp. 1, 7, 10, 15–16
Float/tilt threshold	Binary event	Simple event notification and no continuous sampling requirement	No trend or rate-of-rise information; price and lifetime are not established here	Project-specific device evidence required
Contactless ultrasonic flow	Flow estimate and level at 12 Hong Kong locations	Avoids direct submerged installation in the studied configuration	72.14% average relative error against Manning estimates in that study; results are context-specific	Alshami, pp. 1–4, 11–18
Physicochemical probes	Level, temperature, conductivity, turbidity, or pH	Event and water-matrix information; self-cleaning can extend operation in a defined design	Fouling, calibration, and interpretation are site- and device-specific	Shi, pp. 2–4, 11–12; Li, pp. 1, 3–4, 8–9
UV/Vis and fluorescence	Optical proxies for TSS, COD, BOD, or TOC	Rapid, reagent-free proxy measurements are possible	Matrix effects and local calibration/verification are essential	Lepot, pp. 1–2, 10–11; Carstea, pp. 1, 5–6, 9–10
H ₂ S gas/liquid	Headspace gas or dissolved sulfide	Supports odour, corrosion, and dosing-related monitoring questions	Gas and liquid phases require different sensors, calibration, and maintenance	Pacheco, pp. 1–5, 10–15; Despot, pp. 1–5, 13–16
Telemetry and soft sensors	Remote data and estimated states	Makes distributed monitoring operationally useful	Coverage, power, data quality, and model transfer must be validated	Drenoyanis; Sikorski

10 Implications for a Future SA Water Placement Study

The literature supports a purpose-specific mixed-sensor study, not a fixed technology hierarchy or a universal percentage split. A defensible placement model should make sensor type and location joint decisions. Candidate constraints include:

- monitoring purpose: choke or overflow warning, hydraulic calibration, source identification, compliance, H₂S, or asset condition;
- network topology, pipe geometry, existing sensors, critical assets, and historical events;
- access, confined-space risk, telemetry, power, fouling, calibration, maintenance, and redundancy;
- manufacturer performance data and SA Water procurement or maintenance records for any cost, lifetime, or interval claim.

The final allocation should be an output of the validated model and must sum to 100% if the categories are mutually exclusive. It should not be presented as a literature-proven 5–10%, 20–30%, and 60–75% split. Similarly, no sensor arrangement can be said to “guarantee 100% coverage” without a defined network topology, detection threshold, communications test, and historical-event validation.

A defensible methodology statement is:

We propose a purpose-specific mixed-sensor optimisation model for SA Water. Candidate decisions include sensor type and location, subject to network, budget, access, telemetry, power, maintenance, calibration, and existing-deployment constraints. The model will be validated against SA Water topology, historical choke and overflow records, sensor data, and manufacturer performance data. Literature findings are used as priors and design constraints; the final sensor mix, percentages, lead times, and coverage are project outputs to be tested, not assumed facts.

11 Conclusion

The audited literature supports the following conclusions:

1. Sewer monitoring is best classified by the question being answered: level, flow, water quality, asset condition, or data transmission.
2. Non-contact radar is a credible candidate for level monitoring in humid sewer environments, but Drenoyanis et al. does not establish a generic 77–81 GHz, fifteen-manhole, twelve-month deployment or a five-year battery result.
3. Alshami et al.’s 72.14% value is a context-specific result from contactless ultrasonic sensing and Manning-equation comparison, not a radar result or universal flow-sensing error.
4. Water-quality and optical measurements provide useful event indicators and proxies, but fouling, matrix effects, local calibration, and verification are central limitations.
5. Gas-phase and liquid-phase H₂S measurements must remain separate categories. Their ranges, humidity limits, maintenance needs, and calibration requirements are device- and phase-specific.
6. Costs, battery lifetimes, maintenance intervals, RF penetration, accuracy guarantees, and deployment percentages require primary procurement, manufacturer, or SA Water field evidence before being used in an official recommendation.

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